

RFID Manager

Test Setup — Overview

MQTT Infrastructure • NFC Scanning • End-to-End Testing

2026-06-14

System Overview

Full architecture from NFC scan to data persistence



Android App (RFID Manager)

- NFC Reader/Writer
- MQTT Client (Paho)
- Room DB (local cache)
- Scan → Publish flow
- IP: 192.168.50.112
- WiFi-connected



MQTT Broker (Mosquitto)

- Docker container
- eclipse-mosquitto:latest
- Port 1883 (TCP)
- allow_anonymous: true
- IP: 192.168.50.128
- Mosquitto 2.1.2



Subscriber / Backend

- Python subscriber script
- SQLite persistence
- Filters: rfidmanager/+/telemetry
- Color-coded terminal output
- Statistics on exit
- Localhost connection



Data Flow

1. NFC-tag scanned → UID extracted
2. JSON payload constructed
3. Published to rfidmanager/<uid>/telemetry
4. Broker routes to all subscribers
5. Python subscriber logs to SQLite
6. App stores locally in Room DB



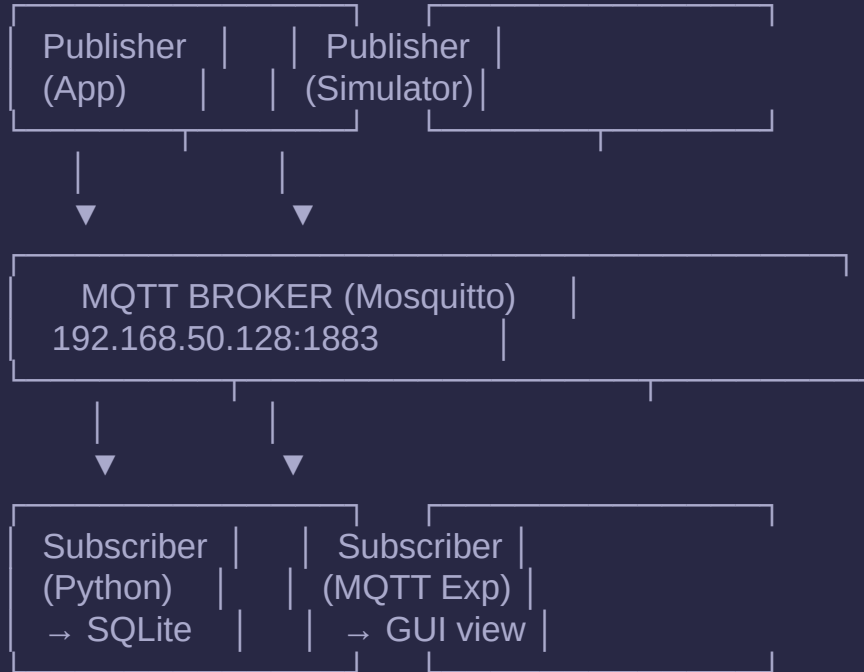
Test Configuration

- Broker: 192.168.50.128:1883
- Topic: rfidmanager/<uid>/telemetry
- QoS: 1 (at least once)
- Keep-alive: 30s
- cleanSession: true
- JSON format payload

MQTT Infrastructure

Publish/Subscribe pattern — decoupled architecture

Publish / Subscribe Model



Key points:

- Publishers don't know subscribers
- Broker handles all routing
- Any number of pub/sub on each topic
- Wildcards enable flexible subscriptions

Topic Structure

rfidmanager / <uid> / telemetry

↑ ↑ ↑
our app tag ID data type

Examples:

- rfidmanager/047B1234/telemetry
- rfidmanager/0A1B2C3D/telemetry

Wildcards:

- rfidmanager/# — all messages
- rfidmanager/+/telemetry — all UUIDs
- rfidmanager/047B1234/+ — all types

JSON Payload:

```
{"uid":"047B1234","type":"rfid",  
  "timestamp":1718300000}
```

QoS Levels:

- 0: fire-and-forget
- 1: at least once (our choice)
- 2: exactly once (overkill)

Network Topology

Home network: 192.168.50.0/24

Phone (WiFi)

- Samsung Galaxy Note 10
- IP: 192.168.50.112
- Paho MQTT Client
- clientId: rfid-android-client
- Publishes on scan

PC — Docker Host

- IP: 192.168.50.128
- Mosquitto container (port 1883)
- Python subscriber
- MQTT Explorer
- All on same machine

External Tools

- mosquitto_pub/sub (standalone or via docker)
- MQTT Explorer (GUI)
- Wireshark (packet analysis)
- netcat (raw TCP test)

Connection Matrix

Component	IP/Host	Connects To	Port	Protocol
Phone (App)	192.168.50.112	Broker (WiFi)	1883	MQTT
Broker (Container)	192.168.50.128	— (accepts all)	1883	TCP
Python Subscriber	localhost	Broker (local)	1883	MQTT
MQTT Explorer	192.168.50.128	Broker	1883	MQTT
mosquitto CLI	localhost/192.168.50.128	Broker	1883	MQTT
Wireshark	any	Network interface	—	MQTT ftr

Test Tools & Commands

Three ways to test — CLI, Docker, or GUI

Method 1: Standalone mosquitto CLI

Install: `sudo pacman -S mosquitto`

Listen on all topics:

```
mosquitto_sub -h 192.168.50.128 -p 1883 \  
-t "rfidmanager/#" -v
```

Publish test message:

```
mosquitto_pub -h 192.168.50.128 -p 1883 \  
-t "rfidmanager/test/telemetry" \  
-m '{"uid":"test"}
```

Method 3: MQTT Explorer (GUI)

Download: github.com/thomasnordquist/MQTT-Explorer

Connection settings:

Host: 192.168.50.128
Port: 1883
SSL/TLS: Off
Auth: None

Subscribe to `rfidmanager/#` for real-time view

Method 2: Via Docker container

No installation needed — broker already has mosquitto tools.

Listen:

```
docker exec rfid-mqtt-test mosquitto_sub \  
-h localhost -p 1883 -t "rfidmanager/#" -v
```

Publish:

```
docker exec rfid-mqtt-test mosquitto_pub \  
-h localhost -p 1883 \  
-t "rfidmanager/test/telemetry" \  
-m '{"type":"test"}
```

Python Test Scripts

Location: `~/rfid-manager/test/fas2-mqtt/mqtt/`

Subscriber (SQLite logging):

```
./test_subscriber_persist.py  
./test_subscriber_persist.py --uid 047B
```

Simulator (simulate phone):

```
./simulate_mobile_publish.py
```

Output: `~/rfid-manager/data/rfid_readings.db`

Broker Administration

Docker-based Mosquitto management

Docker Commands

Command	Description
<code>docker run ... eclipse-mosquitto</code>	Start broker (--rm!)
<code>docker start rfid-mqtt-test</code>	Start existing cont.
<code>docker stop rfid-mqtt-test</code>	Stop broker
<code>docker restart rfid-mqtt-test</code>	Restart broker
<code>docker ps --filter name=rfid-...</code>	Check if running
<code>docker logs rfid-mqtt-test</code>	View logs
<code>docker logs -f rfid-mqtt-test</code>	Follow logs live
<code>docker exec -it rfid-mqtt-test sh</code>	Shell inside container
<code>docker inspect rfid-mqtt-test</code>	Detailed info

⚠ The container uses --rm: it is deleted on stop.
You must run 'docker run' again, not 'docker start'.
Omit --rm if you want persistent container lifecycle.

mosquitto.conf

File: ~/rfid-manager/test/fas2-mqtt/mqtt/mosquitto.conf

Config:

```
listener 1883          # MQTT port
allow_anonymous true   # No auth required
persistence true       # Data saved to disk
persistence_location   # /mosquitto/data/
log_dest stdout        # Logs to console
```

Not configured (defaults):

- max_connections: unlimited
- keepalive_interval: 60s
- allow_zero_length_clientid: true

Logs show:

- New connections (IP + port)
- Client IDs
- Disconnections

End-to-End Test Flow

From NFC scan to data in SQLite — complete chain

1 NFC Scan

Phone reads
RFID tag
→ extracts UID

2 Build Payload

JSON constructed:
uid, type,
timestamp

3 Publish MQTT

→ rfidmanager/
<uid>/telemetry
QoS 1

4 Broker Routes

Mosquitto receives
and forwards
to subscribers

5 Subscriber

Python script
receives →
parses JSON

6 SQLite Store

Data persisted
in rfid_readings.db
for analysis

✓ How to Verify

Method	What to check	Pass	
Terminal: python subscr.	See incoming messages with UID and timestamp	✓	
Terminal: mosquitto_sub	-v flag shows exact topic + payload	✓	
GUI: MQTT Explorer	Real-time tree of all topics	✓	
App: Settings	Broker status indicator	✓	
SQLite: rfid_readings.db	Query: SELECT * FROM readings;	✓	
App: Readings screen	Scan appears in list	✓	

Quick Reference

Essential commands and configurations



Connection Details

Parameter	Value
Broker IP	192.168.50.128
Port	1883
Protocol	MQTT 3.1.1
Topic root	rfidmanager/
QoS	1
Auth	None (anon)
TLS	No
Keep-alive	30s
Clean sess	true



Important Files

Path	Purpose
mosquitto.conf (test/fas2-mqtt/mqtt/)	
test_subscriber_persist.py (test/fas2-mqtt/mqtt/)	
simulate_mobile_publish.py (test/fas2-mqtt/mqtt/)	
rfid_readings.db (data/)	
MQTT-Infrastruktur.md (wiki/)	



One-Liners

Verify broker is up:

```
docker ps | grep mqtt
```

Listen to everything:

```
docker exec rfid-mqtt-test \
  mosquitto_sub \
  -h localhost -p 1883 \
  -t "rfidmanager/#" -v
```

Publish test:

```
docker exec rfid-mqtt-test \
  mosquitto_pub -h localhost \
  -p 1883 -t "test" -m "ping"
```

View logs:

```
docker logs rfid-mqtt-test
```


Questions?

RFID Manager — MQTT Test Infrastructure

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